Discs & hubs - design brief

AI BRAKING / AB-ROT / C0 / 2026-09-07

A 400 mm disc and detachable flanged hub define a shared prototype interface. The drum variant appears in the drum-brake package. This is a geometry study, not a speed-rated rotor.

TARGETS (not validated)

Disc outside diameter: 400 mm. Modelled geometry

Disc thickness: 20 mm. Modelled geometry

Hub shaft bore: 50 mm. Illustrative shaft interface

Mounting pattern: 6 × Ø13 on PCD 150 mm. Modelled; M12 candidate

Torque / speed rating: Not assigned. Shaft, balance and thermal assessment required

DESIGN DECISIONS

Use a detachable hub to replace the friction disc without replacing the shaft interface.

Begin with machined steel for prototype traceability; evaluate a cast-iron friction variant only through paired friction and thermal tests.

Use one six-hole interface across the first disc concepts; bolt preload and joint slip capacity remain to be sized.

CANDIDATE MATERIALS

Disc: C45 normalized steel candidate; friction compound must be qualified with this surface.

Hub: 42CrMo4 quenched-and-tempered candidate; property certificate must correspond to actual section size.

Assembly and engineering gates

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ASSEMBLY INTENT

1. Inspect disc and hub identification, bore and interface faces.

2. Locate disc on the hub spigot; align six through holes.

3. Fit the specified through-bolts, washers and nuts after joint design is approved.

4. Measure assembled runout and balance before guarded testing.

OPEN DESIGN RISKS

1. Disc-to-hub torque transfer, bolt preload and fatigue are not yet established.

2. No keyway or locking element is released; the 50 mm bore is an interface placeholder.

3. Thermal coning, hot runout, balance and overspeed remain unverified.

VALIDATION REQUIRED

1. Thermal transient model with moving pad heat flux and measured heat partition.

2. Bolted joint slip, bearing, hub torsion and rotational stress calculations.

3. Guarded speed, runout, balance and repeated stop tests.

ELECTRICAL / SENSOR REQUIREMENTS ONLY

1. Reserve non-contact runout measurement access and a temperature observation window.

Assembly reference / numbered components

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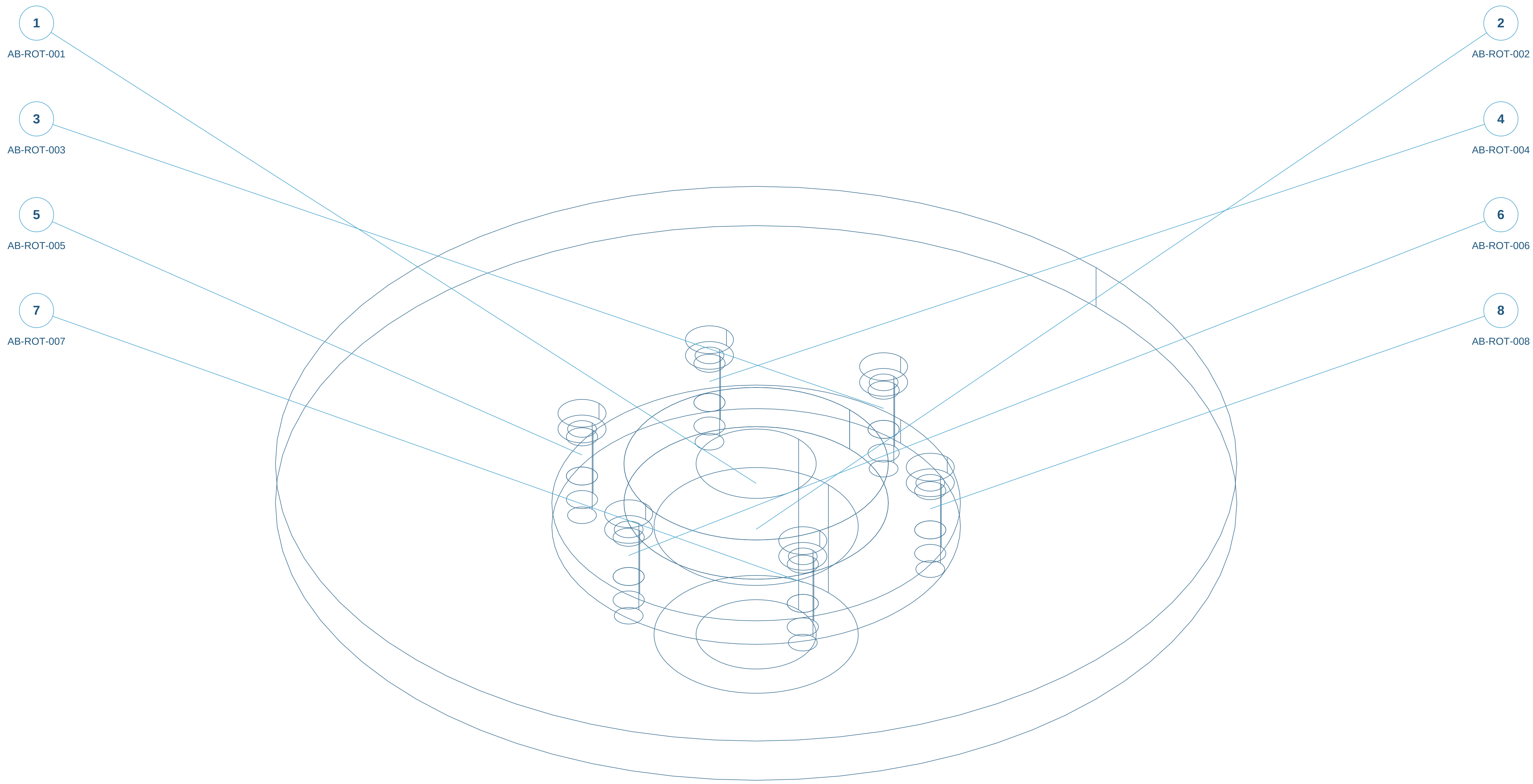

Static isometric reference with item marks. Exploded geometry and part selection are available in the web workspace.

ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

Parts and interfaces

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AB-ROT-001 | Disc 400 | make-concept

C45 normalized candidate; friction pair unqualified; Saw/laser blank; CNC critical faces and bores

AB-ROT-002 | Detachable flanged hub | make-concept

42CrMo4 candidate; heat treatment / fit unapproved; Saw/laser blank; CNC critical faces and bores

AB-ROT-003 | M12 interface fastener envelope 1 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy certified bolt/nut/washer set after joint calculation

AB-ROT-004 | M12 interface fastener envelope 2 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy certified bolt/nut/washer set after joint calculation

AB-ROT-005 | M12 interface fastener envelope 3 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy certified bolt/nut/washer set after joint calculation

AB-ROT-006 | M12 interface fastener envelope 4 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy certified bolt/nut/washer set after joint calculation

AB-ROT-007 | M12 interface fastener envelope 5 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy certified bolt/nut/washer set after joint calculation

AB-ROT-008 | M12 interface fastener envelope 6 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy certified bolt/nut/washer set after joint calculation

AB-ROT-001 / Disc 400

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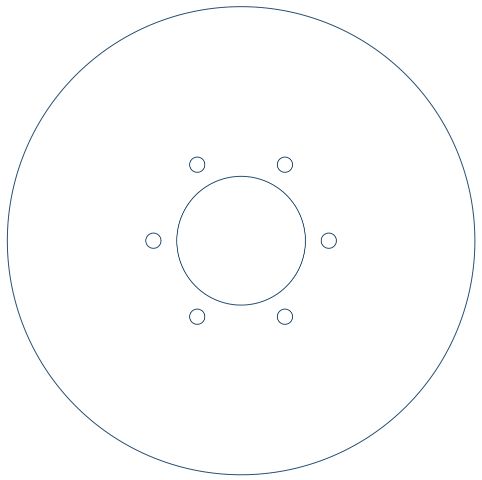
TOP X-Y



C45 normalized candidate; friction pair unqualified

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 400.00 / 400.00 / 20.00 mm

OD 400; bore 110; thickness 20 mm.

6 x diameter 13 THRU on PCD 150; first hole on +X.

Candidate M12 through-bolts; preload and friction joint unresolved.

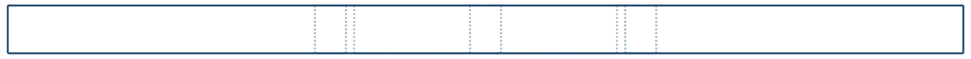
CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z

SIDE Y-Z



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AB-ROT-002 / Detachable flanged hub

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42CrMo4 candidate; heat treatment / fit unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 170.00 / 170.00 / 87.00 mm

Shaft bore 50; spigot OD 110; flange OD 170 mm.

Flange 12 mm; 6 x diameter 13 THRU on PCD 150.

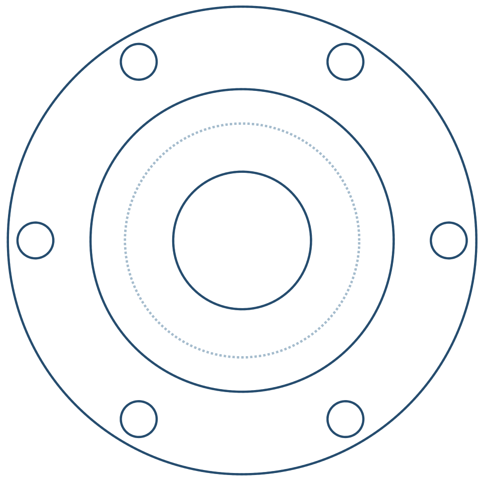
No shaft locking, keyway, fit or retention released.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

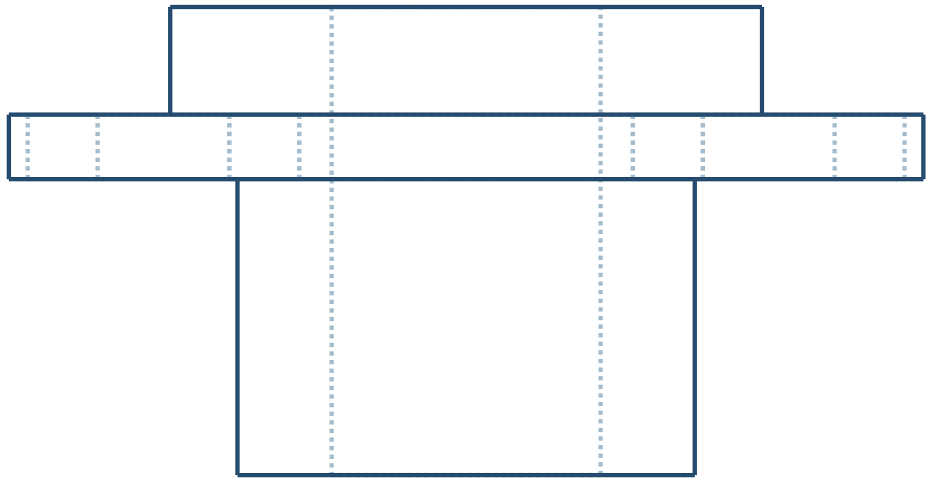
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

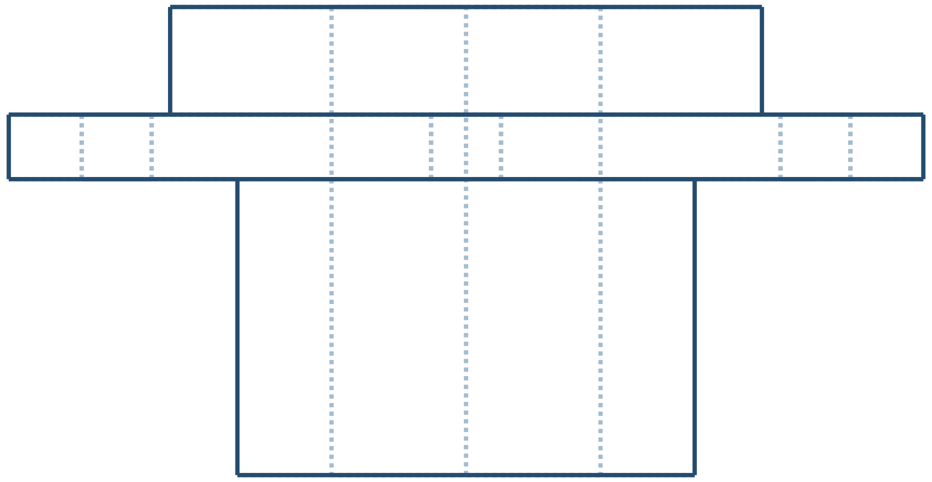
TOP X-Y



FRONT X-Z



SIDE Y-Z



ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

AB-ROT-003 / M12 interface fastener envelope 1

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Supplier-controlled; illustrative reserved envelope

Buy certified bolt/nut/washer set after joint calculation

Overall X/Y/Z bounds: 20.00 / 20.00 / 52.00 mm

Smooth nominal shank only; threads, washer and nut omitted.

Fastener length, grade and preload not released.

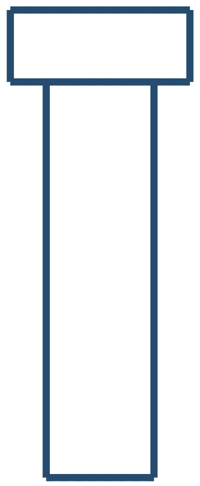
CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

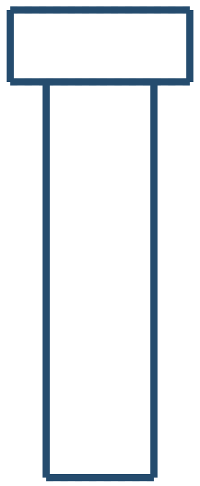
Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

TOP X-Y

FRONT X-Z



SIDE Y-Z



AB-ROT-004 / M12 interface fastener envelope 2

AI BRAKING / AB-ROT / C0 / 2026-09-07

TOP X-Y

Supplier-controlled; illustrative reserved envelope

Buy certified bolt/nut/washer set after joint calculation

Overall X/Y/Z bounds: 20.00 / 20.00 / 52.00 mm

Smooth nominal shank only; threads, washer and nut omitted.

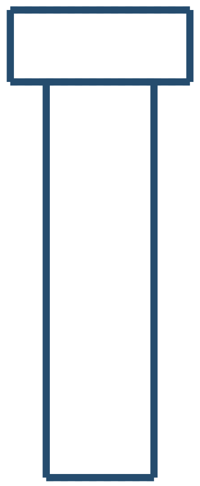
Fastener length, grade and preload not released.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

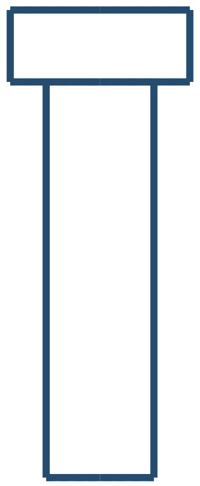
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z



SIDE Y-Z



ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

AB-ROT-005 / M12 interface fastener envelope 3

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TOP X-Y

Supplier-controlled; illustrative reserved envelope

Buy certified bolt/nut/washer set after joint calculation

Overall X/Y/Z bounds: 20.00 / 20.00 / 52.00 mm

Smooth nominal shank only; threads, washer and nut omitted.

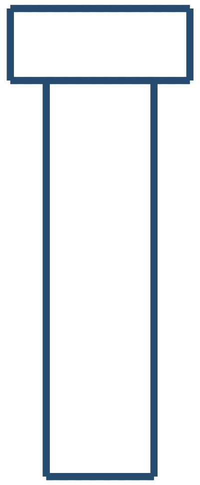
Fastener length, grade and preload not released.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

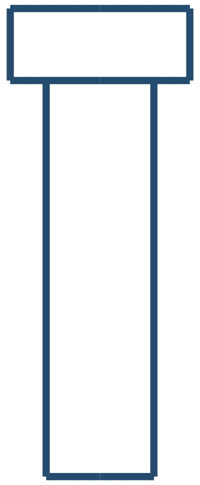
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z



SIDE Y-Z



ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

AB-ROT-006 / M12 interface fastener envelope 4

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TOP X-Y

Supplier-controlled; illustrative reserved envelope

Buy certified bolt/nut/washer set after joint calculation

Overall X/Y/Z bounds: 20.00 / 20.00 / 52.00 mm

Smooth nominal shank only; threads, washer and nut omitted.

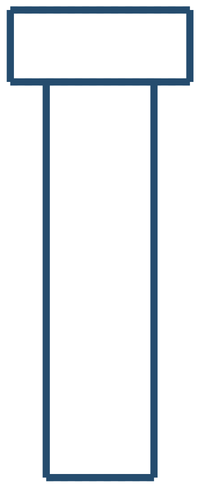
Fastener length, grade and preload not released.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

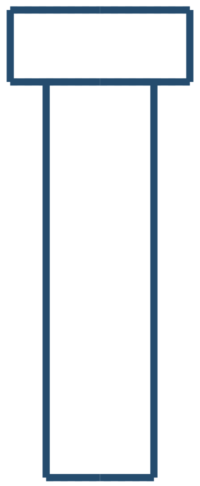
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z



SIDE Y-Z



ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

AB-ROT-007 / M12 interface fastener envelope 5

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Supplier-controlled; illustrative reserved envelope

Buy certified bolt/nut/washer set after joint calculation

Overall X/Y/Z bounds: 20.00 / 20.00 / 52.00 mm

Smooth nominal shank only; threads, washer and nut omitted.

Fastener length, grade and preload not released.

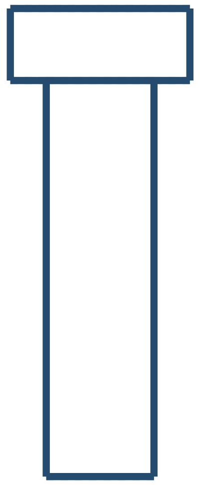
CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

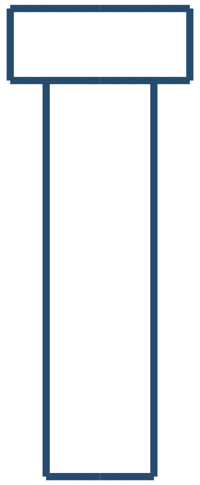
Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

TOP X-Y

FRONT X-Z



SIDE Y-Z



ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

AB-ROT-008 / M12 interface fastener envelope 6

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TOP X-Y

Supplier-controlled; illustrative reserved envelope

Buy certified bolt/nut/washer set after joint calculation

Overall X/Y/Z bounds: 20.00 / 20.00 / 52.00 mm

Smooth nominal shank only; threads, washer and nut omitted.

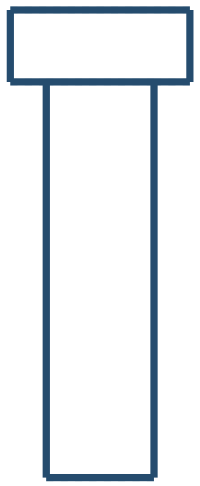
Fastener length, grade and preload not released.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

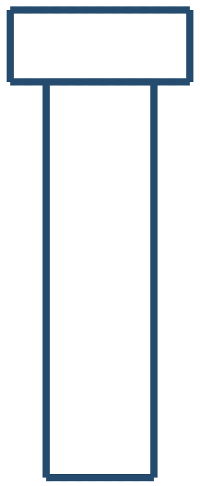
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z



SIDE Y-Z



ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.